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Review

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Chemistry, Nutrition, and Health-Promoting Properties of *Hericium erinaceus* (Lion's Mane) Mushroom Fruiting Bodies and Mycelia and Their Bioactive Compounds

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ABSTRACT: The culinary and medicinal mushroom *Hericium erinaceus* is widely consumed in Asian countries, but apparently not in the United States, for its nutritional and health benefits. To stimulate broader interest in the reported beneficial properties, this overview surveys and consolidates the widely scattered literature on the chemistry (isolation and structural characterization) of polysaccharides and secondary metabolites such as erinacines, hericerins, hericenones, resorcinols, steroids, mono- and diterpenes, and volatile aroma compounds, nutritional composition, food and industrial uses, and exceptional nutritional and health-promoting aspects of *H. erinaceus*. The reported health-promoting properties of the mushroom fruit bodies, mycelia, and bioactive pure compounds include antibiotic, anticarcinogenic, antidiabetic, antifatigue, antihypertensive, antihyperlipodemic, antisenescence, cardioprotective, hepatoprotective, nephroprotective, and neuroprotective properties and improvement of anxiety, cognitive function, and depression. The described anti-inflammatory, antioxidative, and immunostimulating properties in cells, animals, and humans seem to be responsible for the multiple health-promoting properties. A wide range of research advances and techniques are described and evaluated. The collated information and suggestion for further research might facilitate and guide further studies to optimize the use of the whole mushrooms and about 70 characterized actual and potential bioactive secondary metabolites to help prevent or treat human chronic, cognitive, and neurological diseases.

KEYWORDS: *Hericium erinaceus, mushrooms, fruit bodies, mycelia, phylogenetics, nutrients, food use, food processing, bioactive compounds, multifunctional health properties, antioxidant capacity, anti-inflammatory effects, immunostimulation, cell, rodent, and human studies, health-promoting properties, bioactive compounds, erinacines, erinacerins, erinaceolactones, glycoproteins, polysaccharides, resorcinols, sterols, volatile compounds, research needs*

■ INTRODUCTION

Mushrooms are generally a good source of nutrients and health-promoting compounds.[1−3] More than 2000 species of edible and/or medicinal mushrooms have been identified.[4] Among these, the culinary and medicinal *Hericium erinaceus*, also known as Yamabushitake or Lion's Mane, the subject of this review, seems to stand out as an exceptional health-promoting species. This species has the potential to help prevent, alleviate, or treat major diseases, including cancer, depression, diabetes, lipidemia, and neurodegenerative diseases.[5−9] Here, we emphasize the chemical basis for the reported beneficial effects of fruiting bodies and mycelia of this culinary−medicinal mushroom, which is widely consumed in Asian countries (China, India, Japan, and Korea) but apparently not in the United States and elsewhere, in relation to its content of characterized bioactive compounds. A better understanding of the nutritional value in terms of amino acid, protein, carbohydrate, fatty acid, vitamin, and mineral contents and the chemical basis of the antioxidant, anti-inflammatory, and immunostimulating properties that might be responsible for the reported multiple health-promoting properties of whole mushrooms and individual bioactive components is expected to facilitate the development of health-promoting diets supplemented with bioactive mushroom fruiting bodies, mushroom mycelia, selected bioactive fractions, and pure mushroom components. It is hoped that this review will also promote translation of this basic scientific information to applications in nutrition and medicine to help prevent, alleviate, and treat major diseases, including atherosclerosis, cancer, diabetes, and diseases of the heart and the nervous system. This review will contribute to our understanding of the relationship between the composition of mushrooms and their bioactivity in cells, animals, and humans.

■ PHYLOGENETICS OF *HERICIUM* SPECIES

Phylogenetic analysis has been used to differentiate Chinese *H. erinaceus* from related mushrooms. For example, rDNA internal transcribed tracer (ITS) sequences have been used to classify edible and medicinal mushroom species using a time-effective and low-cost PCR method.[10,11] Indeed, Hallenberg et al. successfully used this method to discover a new *Hericium* species.[12] In addition, on the basis of a Bayesian phylogram and diagnostic characteristics of a number of investigated *Hericium* species, the same authors note that *H. erinaceus* appears as three subclades, that the specimens within a subclade generally belong to the same or adjacent geographical regions, that one representative subclade features all of the representative from the United States that grow on *Quercus*, and that another

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subclade is mainly composed of representatives from East Asia. Because new *Hericium* species continue to be discovered,[13] it would be of interest to compare the compositions and bioactivities of taxonomically and phylogenetically related and different *Hericium* species to those of *H. erinaceus*, the subject of the present study.

■ COMPOSITION

H. erinaceus fruit bodies and mycelia contain an exceptionally large amount of structurally different bioactive and potentially bioactive components. Figure 1 shows a photograph of an *H. erinaceus* mushroom with its mane-like surface. Figures 2−6 depict structures arbitrarily subdivided into five classes of organic compounds (erinacines, aromatic compounds, steroids, alkaloids, and lactones) isolated from the fruit bodies and mycelia. It seems to be a unique event in nature for a single mushroom species to biosynthesize about 70 different secondary metabolites. Here, the chemical nature of some of the compounds that might be responsible for the observed health-promoting properties is described below. The reader is referred to the cited references for details about structural and stereochemical characterization and nomenclature.

Figure 1. Photograph of Lion's Mane mushroom (from Congaree National Park, South Carolina, USA, courtesy of the National Park Service).

Hericerins. Hericerins—aromatic compounds with medicinal properties—have been isolated and identified from *H. erinaceus*. In a recent publication, Li et al. describe the isolation and structural characterization of two new aromatic compounds (hericerin A and isohericenone J) along with five known compounds (isoericerin, hericerin, *N*-dephenylethyl isohericerin, hericenone J, and 4-(3′,7′-dimethyl-2′,6′-octadienyl)-2-formyl-3-hydroxy-5-methoxybenzylalcohol) from a chloroform extract of the *H. erinaceus* fruiting bodies.[14] In related studies, Li et al.[15] describe the isolation of a new aromatic compound, erinacene D, from *H. erinaceus*. Miyazawa et al.[16] reported on the isolation of two new aromatic compounds, including a new D-arabinol ester, and two known compounds from *H. erinaceus*, and Zhang et al.[17] identified a new diterpene from *H. erinaceus* mycelia. The anticancer properties of some of these compounds are mentioned below.

Kobayashi et al.[18] achieved the total synthesis of bioactive resorcinols, erinacerins, and hericenols, isolated from the fruiting bodies of *H. erinaceus*. The structure of hericenone B was revised as the carbonyl regioisomer at the lactam moiety.

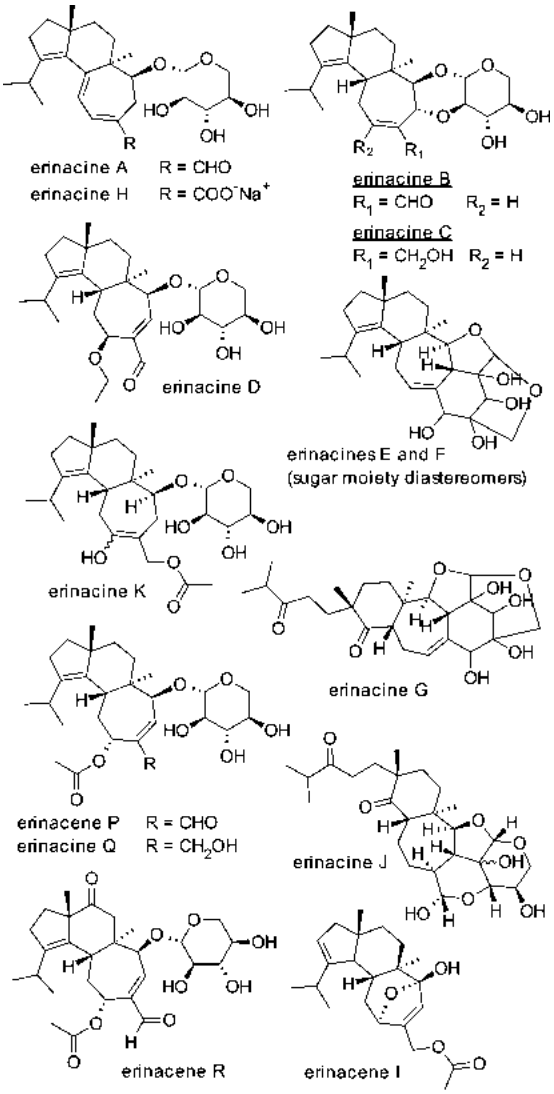

Figure 2. Structures of erinacines isolated from *Hericium erinaceus* fruit bodies and mycelia. Researchers at Shizuoka University first identified erinacines A−I;[80,103,132−134] Kenmoku et al.[135,136] characterized erinacines P and Q; and Ma et al.[137] characterized erinacine R.

Erinacines. The erinacines are a group of cyathane-type diterpenoids, including 20 members of 24 diterpenoids in *H. erinaceus* described by Tang et al.,[19] that have potential neuroprotective properties. Lee et al.[20] isolated erinacine A from an ethanol extract *H. erinaceus* mycelia. The compound contains the usual seven-member ring with two conjugated double bonds, which enables analysis by HPLC with UV detection at 340 nm.

Erinacerins—Isoindolin-1-ones. The isoindoline-1-ones are another group of secondary metabolites produced by *H. erinaceus*. Wang et al.[21] isolated 10 new isoindoline-1-ones, named erinacerins C−L, from a solid *H. erinaceus* culture. The structures of the compounds were established using spectroscopic methods, and the absolute configurations of some of these were assigned by comparing the specific rotations with those of related phthalimidines. The authors suggest that the described solid-state fermentation technique seems to be a useful method to stimulate the biosynthesis of secondary metabolites in edible and medicinal mushrooms and offer a hypothetical biogenesis pathway for the formation of the

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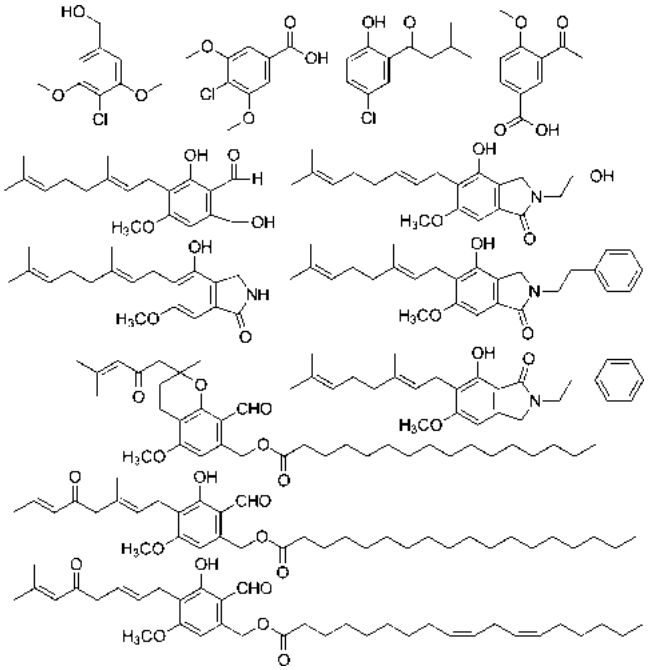

Figure 3. Structures of aromatic compounds isolated from *Hericium erinaceus* fruit bodies and mycelia. Adapted from Li et al.,[14] Noh et al.,[109] Zhang et al.,[17] and Miyazawa et al.[16]

erinacerins. Some of the compounds inhibited the growth of cancer cells, as described below.

Erinaceolactones. A recent study describes the isolation and characterization by spectroscopic methods of three novel erinaceolactones from the culture broth of *H. erinaceus*.[22] The absolute configuration of one of the compounds was determined by X-ray crystallography. A bioassay of plant-growth regulatory activity revealed that the new and structurally related *Hericium* compounds inhibited hypocotyl and root growth of lettuce at nanomolar concentrations.

Glycoprotein. Cui et al. isolated a novel glycoprotein (HEG-5) from cultures of *H. erinaceus* mycelia. The compound has a molecular weight of 14.4 kDa and has been shown to have hemagglutinating activity.[23] The protein/polysaccharide ratio of HEG-5 is 10:1 (%/%), and it contains D-glucose, L-rhamnose, D-galactose, and D-mannose in a molar ratio of 1.00:1.09:2.45:7.14 in the polysaccharide fraction. Analysis using FT-IR and NMR showed that HEG-5 contains the protein and carbohydrate parts with (1→4)-linked β-galactose and β-glucose residues, and circular dichroism showed that the HEG-5 is a predominantly β-sheet glycoprotein. The complete three-dimensional structure was obtained by protein sequencing and modeling using matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF-MS), NCBI blast search, and online SWISS-MODEL Workplace service. The glycoprotein inhibited the growth of human gastric carcinoma cells in the MTT cell viability assay and exhibited hemagglutinating, but not hemolytic, activity in rat erythrocytes, as described below.

Polysaccharides. Polysaccharides from *H. erinaceus* have been of particular interest because of the bioactivities that have been attributed to them. There are therefore a number of studies on their isolation and analysis. For example, a Russian study[24] describes the isolation of two polysaccharides from *H. erinaceus* mycelia that differ in solubility in aqueous ethanol. Both products contained the following sugars in different ratios: fucose, galactose, glucose, rhamnose, and xylose.

Wu et al. used colorimetry with iodine and KI, high-performance size exclusion chromatography coupled with multiangle laser light scattering and refractive index detection (HPSEC-MALLS-rid analysis), gas chromatography−mass spectrometry (GC-MS) analysis, and saccharide mapping based on carbohydrate gel electrophoresis (PACE) analysis to identify polysaccharides isolated from *H. erinaceus* fruiting

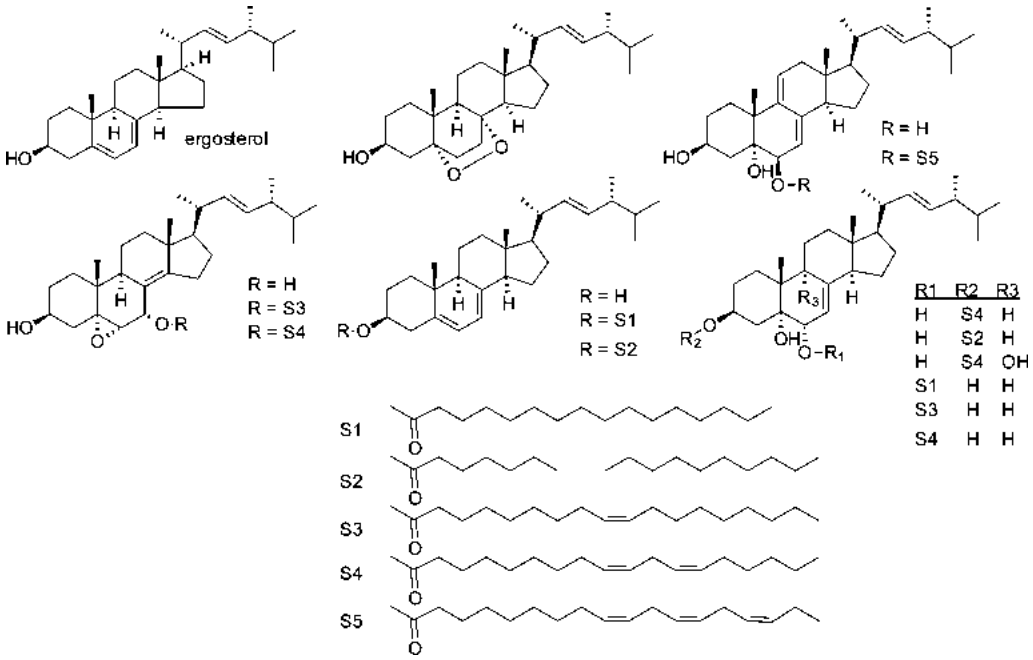

Figure 4. Structures of steroid compounds isolated from *Hericium erinaceus* fruit bodies and mycelia. Adapted from Avtonomova et al.,[24] Zan et al.,[32] and Li et al.[44]

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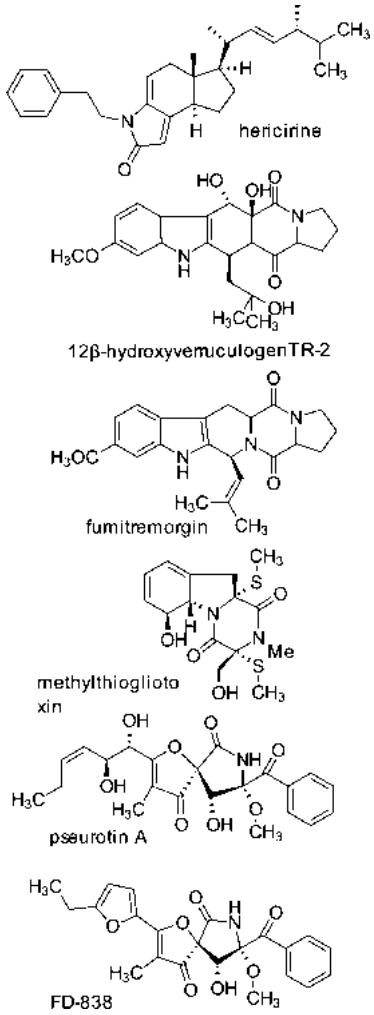

Figure 5. Structures of alkaloids isolated from *Hericium erinaceus* fruit bodies and mycelia. Adapted from Li et al.[138] and Lu et al.[139]

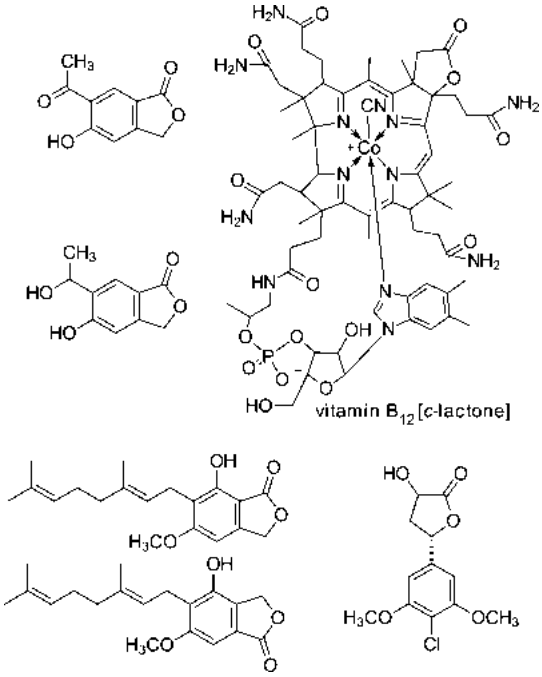

Figure 6. Structures of lactones isolated from *Hericium erinaceus* fruit bodies and mycelia. Adapted from Wu et al.[22] and Teng et al.[47]

bodies collected from different parts of China.[25] The results show similarity in their profiles, including molecular weights, the composition of monosaccharides, and the glycosidic linkages in the polysaccharides, suggesting that the described experimental approach could be used for quality control of polysaccharides in edible and medicinal mushrooms as well as in commercial mushroom products. In addition, Hou et al.[26] isolated a water-soluble oligosaccharide from *H. erinaceus* with antioxidative properties. Other studies on the identification and characterization of *H. erinaceus* polysaccharides (β-D-glucans) are also reported by Kim et al.,[7] Bhandari et al.,[27] Li et al.,[28] Malinowska et al.,[29] Semedo et al.,[30] Shang et al.,[31] Wu et al.,[25] Zan et al.,[32] Zhang et al.,[33,34] and Zhu et al.,[35] some of which we elaborate here.

Using HPSEC to establish the degree of homogeneity and molecular weights, Malinowska et al.[29] evaluated several factors that govern the mycelial growth in a submerged culture of *H. erinaceus* and production of intracellular polysaccharides (IPS) and exopolysaccharides (EPS). They found that the kinetics of EPS biosynthesis in a bioreactor proceeded until depletion of the carbon source in the medium on day 14 of cultivation, suggesting the value of the method for large-scale production of bioactive polysaccharides from *H. erinaceus*.

An enzyme mixture (cellulase, pectinase, and tryptinase) facilitated extraction of polysaccharides from the fruits of *H. erinaceus* mushrooms using response surface methodology and the Box−Behnken design based on single-factor and orthogonal experiments to optimize the extraction conditions (pH 5.7, 52.0 °C, and 33.8 min).[35] The optimal extraction produces a high yield (13.5%) corresponding to an increase of 67.7% compared to hot water extraction. The functional groups of the two extracted polysaccharides seemed to be identical but differed in conformations. The authors suggest that the method might be useful for extracting polysaccharides from other mushroom species. Related studies describe the isolation and structural characterization of a rhamnoglucoagalactan[36] and a β-D-glucan[37] from *H. erinaceus* fruiting bodies.

H. erinaceus and other mushroom varieties are reported to be a useful source of lignocellulose-degrading (ligninolytic) carboxymethyl cellulase and laccase enzymes, suggesting that mushrooms can provide enzymes for use in the development of bioenergy.[38] Mushroom spent straw, an important lignocellulosic biomass, might serve as a potential substrate for an ethanol-based biorefinery.[39]

Sterols. The identification of various sterols from *H. erinaceus* has also been reported. For example, Avtonomova et al.[24] detected the presence of ergosterol in the *H. erinaceus* mycelia biomass. Because exposure of ergosterol to UV(B) light transforms ergosterol in mushrooms to vitamin D,[40,41] a process used to commercially market vitamin D-containing mushrooms, it would be of interest to find out whether such exposure would also induce an analogous transformation of *H. erinaceus* fruiting bodies and mycelia, thus further enhancing the health-promoting potential of this mushroom.

Because the amounts of such bioactive sterols, including ergosterol, are fairly low in *H. erinaceus*, Dai et al. evaluated the biomass and ergosterol content of *H. erinaceus* mycelium following supplementation of the liquid culture with the plant-derived compounds salicylic acid and methyl jasmonate.[42] The highest ergosterol content of 2.33 mg/g was obtained following 6 days of cultivation with 100 μmol/L salicylic acid. With 50

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μm/L methyl jasmonate, the highest ergosterol content was 2.0 mg/g. Both values were significantly higher than ergosterol production without the additives. Because methyl jasmonate induced significant increases in both mycelium biomass and ergosterol content during the logarithmic growth phase, whereas salicylic acid enhanced only the production of ergosterol but not biomass, the authors conclude that the use of added jasmonate might be preferable over methyl salicylate.

In other examples of sterol identification, Li et al.[43] compared the nature of sterols present in *H. erinaceus* in ethanol and water extracts, finding them mainly in the ethanol extract and identifying new sterols. Related studies describe the isolation and structural characterization of 6 new (erinarols A−F) and 5 known ergostane-type sterol fatty acid esters and 10 known ergostane-type sterols and erinarols G−J from methanol extracts of the dried fruiting bodies of *H. erinaceum* as well as anti-inflammatory and antiproliferative properties of some of these compounds in cell assays.[32,44]

Vitamin B_{12} Lactone. Some mushrooms, not including *H. erinaceus*, are reported to contain vitamin B_{12},[45] which is of particular interest to vegetarians,[46] because animal-derived foods are the usual dietary sources of vitamin B_{12}. *H. erinaceus* is, however, reported to contain the biologically inactive vitamin B_{12}[c-lactone],[47] and it may be possible to generate the active vitamin from this compound by chemically opening the lactone ring.

Volatile Compounds. Analysis of dried fruiting bodies of *H. erinaceus* by GC and GC-MS identified 77 components, representing 87.6% of the total volatile oil.[48] The main components of the oil were hexadecanoic acid (26.0%), linoleic acid (13.1%), phenylacetaldehyde (8.9%), and benzaldehyde (2.5%). Aroma evaluation of the oil using GC-MS−olfactometry and aroma extract dilution analysis showed that 2-methyl-3-furanthiol, 2-ethylpyrazine, and 2,6-diethylpyrazine seem to be the major contributors to the mushroom aroma. Because plant essential oils and aromatic aldehydes inactivate pathogenic bacteria,[49,50] will the mushroom oil and components also be active against pathogenic microorganisms? (See also Antibiotic Properties.)

■ NUTRIENTS, NUTRITION, AND FOOD USE

Here, we will first mention some general aspects associated with mushroom production, composition, and nutrition followed by a discussion of the limited studies on the nutrition of *H. erinaceus*.

Nutritional Composition. In describing culinary aspects of mushrooms, McGee notes that mushrooms are fungi that have no chlorophyll and thus they cannot harvest energy from the sun.[51] Instead, they live off plants and plant remains. They are therefore distinct from biologically derived foods such as cereals, fruits, and vegetables, and meat and dairy products.[1] The distinctive flavor of mushrooms is associated with a high content of free amino acids, especially glutamic acid, and numerous volatile compounds. Many mushrooms contain enzymes such as polyphenol oxidase that catalyze rapid enzymatic browning reactions[52] and, therefore, darken rapidly when cut or crushed.

A comprehensive review by Sawant et al.[53] provides information on mushroom cultivation and production methods including solid state and submerged culture; diseases caused by pathogenic fungi, bacteria, viruses, and pests; harvesting, grading, and packaging; storage; and processing (canning, dehydration, freezing and freeze-drying, and pickling). The reader is referred to the above-mentioned chapter by Savant for specific details on growing, harvesting, and processing of mushrooms.

An earlier review by Kurtzman,[54] reports that (a) the proximate composition of five mushroom species (not including *Hericium*) ranged as follows: water, from 73 to 91%; crude protein (N × 4.38), from 10.5 to 28.5%; fat, from 1.6 to 8.0%; carbohydrate, from 57.4 to 81.8 g/100 g dry wt; fiber, from 7.5 to 17.4 g/100 g dry wt; ash, from 6.1 to 12.0 g/100 g dry wt; and energy, from 271 to 387 kcal/100 g dry weight; and (b) the total essential amino acid composition ranged from 38.4 to 50.1 g amino acid/100 g of crude protein compared to 51.3 g amino acid/100 g for high-quality egg protein, suggesting that the nutritional quality of these mushrooms is quite high and that the content of essential amino acids approaches that of egg protein.

Ulziijargal and Mau summarize the results of proximate composition studies of 38 fruiting bodies and 19 mycelia of 32 species of culinary−medicinal mushrooms, including *Hericium*.[55] Most fruiting bodies and mycelia are low in fat and rich in protein and dietary fiber; some are rich in polysaccharides and others in crude fiber, suggesting that proximate composition and energy for fruiting bodies and mycelia should be taken into account for proposed uses as culinary and health food and as food additives.

In another study, Rodrigues et al. determined the composition and nutritive values of five mushroom species, including *H. erinaceus*.[56] Protein, sugar, and fat contents were in the ranges of 16.22−26.6, 52.7−64.9, and 2.3−3.5 g/100 g dry mushrooms, respectively. These mushroom are also good sources of phenolic compounds and contain more monounsaturated and polyunsaturated fatty acids than saturated fatty acids, with a high concentration of linoleic acid (>30 g/100 g fat), and they are good sources of magnesium, phosphorus, and especially potassium (2.6−3.7 g/kg dry mushroom). The mushrooms also contain significant amounts of copper, iron, and zinc, suggesting their dietary nutritional potential as a source of multiple macro- and micronutrients.

A comprehensive quantitative study by Cohen at al. of the composition of 15 Basidiomycetes medicinal mushroom strains belonging to 12 species, including *H. erinaceus* fruiting body (FB) and mycelia biomass (MB), found that almost all investigated mushrooms are a good source of proteins and carbohydrates, the levels of which are 8.6−42.5 and 42.9−83.6%, respectively, and that different species exhibited distinct free amino acid profiles.[2] The following information is limited to their reported findings with *H. erinaceus*. The analyses were performed on mushrooms dried at 50 °C and then ground into a powder. The dried powder of *H. erinaceus* FB contained 20.8% protein, 61.1% total carbohydrate, 5.1% fat, 6.8% ash, 6.2% water, and 374 kcal/100 g energy, whereas the MB contained 42.5% protein, 42.9% total carbohydrate 6.3% fat, 4.4% ash, 3.9% water, and 398 kcal/100 energy. The corresponding values for the total free amino acid content are 14.3 and 30.6 mg/g dry weight basis for the FB and MB, respectively. The total fatty acid content consisted of 11 individual unsaturated and saturated forms for FB and 7 for MB. Both FB and MB also contained various amounts of nutritional macro- and microelements (calcium, chromium, cobalt, copper, iron, magnesium, manganese, molybdenum, phosphorus, selenium, sodium, sulfur, and zinc) as well as low amounts potentially toxic elements (aluminum, arsenic, barium, boron, cadmium, nickel, silver, strontium, tin, titanium, and

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vanadium). The content of amino-acid-derived bioactive compounds (in μg/g dry weight) in FB and MB, respectively, was as follows: γ-aminobutyric acid (GABA), 42.9 and 56.0; ergothioneine, 630.0 and 149.2; and lovostatin, 14.4 (FB). A levostatin value for MB was not reported.

It seems that both FB and MB contained useful nutrients and bioactive compounds and can as well serve as a resource for their use in food and pharmaceuticals (neutraceuticals); because of differences in composition among the MB and FB of the same species and among different species, combination of different species could be used to optimize their nutritional and health-promoting potential.

Because the use of mushrooms in main entrées in plant-based diets is growing, Feeney et al.[1,3] suggest that the nutritional and health benefits of mushrooms in terms of their nutrient, culinary, and health-promoting characteristics should be re-evaluated in the context of three separate food kingdoms: plants/botany, animals/zoology, and fungi/mycology. Because of the increasing availability of complete compositional data for different mushroom species, Feeney et al.[1,3] suggest the need for mathematical modeling designed to optimize the nutritional and health potential of combinations of different fruiting bodies and mycelia followed by animal and human feeding studies to confirm the predicted benefits. It may be possible to help meet this need by applying a mathematical modeling method to the relationship of composition to nutritional and health properties of mushrooms developed by Finotti et al.[57,58] for other food categories as well as principal component analysis of the bioactive *H. erinaceus* compounds as described for other mushroom species.[59,60] Moreover, the composition of the mushrooms also seems to depend on whether they are cultivated or grown in the wild.[61]

There is also a need to confirm the calculated nutritional values (scores) of mushrooms based on the content of essential amino acids,[62] by nutritional evaluation using feeding studies with mice, rats, and humans (as described elsewhere for free amino acids and other food proteins).[63−66] Unlike the in vitro results, the in vivo results are influenced by the digestibility of the proteins, absorption of the digested amino acids into the circulation, and their utilization for the synthesis of essential proteins.

Food Uses. The following three studies suggest the potential of *H. erinaceus* as a source of bioactive compounds and antioxidants in bread and in fermented mushroom wine, vinegar, and nonalcoholic beverages. Possible ways to inhibit enzymatic browning in mushrooms are also mentioned.

Bread Additive. Bread baked with 5% of *H. erinaceus* and other mushroom mycelia contained significant amounts of γ-aminobutyric acid (0.23−86 and 0.79−2.1 mg/g dry matter, respectively), reported to affect brain function,[67] and ergothioneine, reported to impact heart function,[68,69] suggesting that adding mushroom mycelia to bread might provide health benefits.[70]

Mushroom Wine and Vinegar. Another food application might be the reported use of *H. erinaceus* as a fermentation substrate for the production of wine and vinegar.[71] After 9 days of fermentation of 5% *H. erinaceus* powder, the alcohol content of the resulting mushroom wine reached 16%. A sensory evaluation by a human panel expressed overall acceptance of the organoleptic properties of the new wine. The wine was then treated with *Acidophilus aceti* bacteria to produce vinegar with an acidity of 4.09% due to presence of acetic, malic, and succinic acids. The new vinegar showed antioxidant activity possibly associated with its phenolic content comparable to those of known vinegars, suggesting the possibility that the *H. erinaceus* vinegar might represent a new food product with antioxidant capacity.

Fermented Mushroom Beverages. Because non-alcoholic fermented plant beverages are produced by fermentation of plant sources with different lactic acid bacteria, Woraharn et al.[72] evaluated the production of L-glutamic and γ-aminobutyric acid in a fermented *H. erinaceus* beverage by two *Lactobacilli* starter cultures. Glutamic acid production was significantly influenced by K_2HPO_4, pH, and temperature, whereas γ-aminobutyric acid production was significantly influenced by pH and temperature, suggesting that the mushrooms can be used to prepare high-quality fermented beverages at low cost.

Inhibition of Browning. Because heat used in cooking and baking is reported to adversely and beneficially affect the composition of mushrooms,[73,74] it would be of interest to find out to what extent food processing (canning, cooking, baking, frying, grilling, microwaving, sautéing) might affect the composition of *H. erinaceus* fruit bodies and mycelia. For example, the enzymatic browning reaction (formation of brown spots) that occurs on storage and heat drying of mushrooms, catalyzed by the enzyme tyrosinase, is a problem that has been addressed by Li et al.[75] for the culinary mushroom *Lentinus edodes* (Shiitake). They pretreated the mushrooms with the amino acid L-cysteine and other inhibitors of the enzyme tyrosinase to inhibit the browning process. L-Cysteine is reported to also inhibit enzymatic browning in potatoes[76] and nonenzymatic (Maillard) browning that can produce the potentially toxic acrylamide.[77,78]

MEDICINAL PROPERTIES

As mentioned, *H. erinaceus* has long been used for its beneficial health properties, which are reported to be wide ranging, as outlined in the sections below. Many of the activities reported for *H. erinaceus* can have crossover (overlapping) effects on different medical conditions as indicated in some of the sections below.

ANTIBIOTIC PROPERTIES

The following observations indicate that *H. erinaceus* extracts and bioactive erinacines and polysaccharides have been shown to have antimicrobial activities against both antibiotic-resistant and nonresistant (susceptible) pathogenic bacteria.

Methicillin-Resistant *Staphylococcus aureus* (MRSA). The prevalence of antibiotic-resistant pathogenic bacteria, especially MRSA, is a major unsolved problem in microbial food safety and human health, as reviewed by Friedman.[79] The identification of new compounds to treat antibiotic-resistant bacteria is therefore welcome. To contribute to this aim, using a Petri dish agar and microplate assays, Kawagishi et al. found that several erinacines isolated from cultured *H. erinaceus* mycelia showed potent anti-MRSA activity, with minimum inhibitory concentrations (MIC) values ranging from 62.5 to 500 μM.[80] An examination of the structure−antimicrobial activity relationship showed that the ring skeleton of the aglycone in the active compounds was indispensable to the anti-MSRA activity. It would be of interest to confirm this useful finding against MRSA-containing food and MRSA-infected animals and humans.[81]

***Salmonella* Typhimurium.** *Salmonella* Typhimurium is a pathogen that contaminates numerous foods, including leafy

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greens, meat, and poultry. Kim et al.[82] discovered that although *H. erinaceus* extracts were inactive against this pathogen in vitro, they protected mice against lethal infection through the activation of innate macrophage immune cells. Two of the four extracts evaluated stimulated the activity of macrophage cells in vitro and protected infected mice against mortality and liver damage. Because systemic salmonellosis induces liver necrosis, liver injury was used as a biomarker for the incidence for in vivo salmonellosis in the mice. It seems that the protection of the liver by the mushroom extracts might be the result of stimulation of the innate immune cells, which then kill the *Salmonella* bacteria that induced the liver injury, analogous to that observed with the cultured macrophage cells. This first report on the inhibition of adverse effects of *Salmonella* in mice via stimulation of the immune system by mushroom extracts suggests that *H. erinaceus* mushrooms or extracts could be used to improve microbial food safety and infectious diseases. It would be of interest to find out whether the *H. erinaceus* extracts and/or pure compounds would also protect mice and humans against adverse effects caused by other foodborne pathogenic bacteria, including *Bacillus cereus*, *Escherichia coli*, *Listeria monocytogenes*, and *Vibrio cholerae*.

In a related study Kim et al.[83] also reported that a polysaccharide isolated from the liquid culture of *L. edodes* mushroom mycelia containing black rice bran protected mice against salmonellosis-induced liver injury and mortality through stimulation of the immune system. The same polysaccharide–black rice bran formulation also protected mice against *Salmonella* lipopolysaccharide-induced endotoxemia.[84] It would be interesting to investigate whether a similar *H. erinaceus* formulation might also protect mice against *Salmonella*-induced lethal infections and often fatal endotoxemia.

Helicobacter pylori. *H. pylori* is a human gastrointestinal pathogen that may contribute to the cause of adverse effects, possibly including ulcers. Shang et al.[85] determined the minimum inhibitory concentration (MIC) values of extracts prepared from the fruiting bodies of 14 mushroom species including *H. erinaceus* against *H. pylori* clinical strains using a microplate agar diffusion assay. MIC values of ethyl acetate fractions of *H. erinaceus* against nine clinical isolates of *H. pylori* ranged between 62.5 and 250 μg/mL. Ethanol extracts were less effective. The extracts of *H. erinaceus* also inhibited *Staphylococcus aureus* (MIC = 10 mg/mL) but had no effect on the growth of two *Escherichia coli* test strains (MIC > 10 mg/mL). The in vitro inhibitory effect of *H. erinaceus* extracts against *H. pylori* suggests antimicrobial mechanisms that do not seem to involve the immunomodulating effects observed with polysaccharides.[82]

Anti-*H. pylori* activities in vitro were evaluated using an agar dilution assay of two novel polysaccharides isolated from *H. erinaceus* with molecular weights of 197 and 20 kDa.[86] Although the polysaccharides alone inhibited the bacteria, complexes with bismuth exhibited stronger inhibition effects (MIC = 20 μg/mL), suggesting the need to evaluate these novel *H. erinaceus*-derived compounds against *H. pylori* infections in humans. It would be of interest to determine whether *H. erinaceus* and bioactive compounds would also inhibit the biological activity of microbial and plant toxins.[87]

Multiple Pathogens. A study by Wong et al.[88] using the well diffusion method that measures inhibition zones on Petri plates found that various *H. erinaceus* extracts inhibited the growth of multiple Gram-positive and Gram-negative pathogenic bacteria. The inhibition depended on the type of extract tested. Oven-dried fruit body extract was inactive.

■ ANTICARCINOGENIC PROPERTIES

Carcinogenesis is a sequential multistage cellular process consisting of tumor initiation, tumor promotion, and tumor progression. Plant compounds in some foods are a potential source of bioactive safe compounds for cancer chemoprevention and suppression of tumor initiation, promotion, and metastasis.[89] Dietary components can protect against cell proliferation and induce cell death (apoptosis) by a variety of mechanisms, including destruction of reactive oxygen species (ROS) that damage DNA and essential proteins, suppression of signaling pathways, and expression of proteins involved in carcinogenesis, inhibition of angiogenesis (blood flow to cancer cells), cancer cell metastasis (migration of cells from the origin to other organs), and stimulation of the immune system that helps eliminate carcinogens.[82,83,90,91] Here, we will briefly mention selected reported studies on the antioxidative, antimutagenic, antitumor-promoting, and anticarcinogenic activities of *H. erinaceus* and bioactive components in mammalian cells and rodents (mice, rats).

Inhibition of Cancer Cells. The recently isolated newly discovered diterpene mentioned under Nutritional Composition above both exhibited anti-*H. pylori* activity and inhibited the growth of human leukemia (K562) and human prostate cancer (LANCAP) cells.[17] The isolated hericerin A and hericerin also significantly reduced the proliferation of leukemia cells with IC_{50} values of 3.06 and 5.47 μM.[14]

A study by Lee and Hong[92] on the mechanism of *H. erinaceus* inhibition of human liver (HepG2) cancer cells showed that (a) hot-water *H. erinaceus* extract enhances the apoptosis induced by the clinical cancer drug doxorubicin and (b) the *H. erinaceus* extract acts as an enhancer to sensitize doxorubicin-mediated apoptotic signaling by reducing C-FLIP expression via JNK activation and enhancing intracellular doxorubicin accumulation via inhibition of NF-κB activity. The mushroom extract seems to have the potential to provide therapeutic effects against drug-resistant human liver carcinoma.

Li et al. isolated a lectin (agglutinin) from dried fruiting bodies of *H. erinaceus* with a molecular mass estimated at 51 kDa, a hemagglutinating activity of 12.5 mM by the inulin assay, and high heat and pH stability.[93] The lectin showed antiproliferative activity toward hepatoma (HepG2) and breast cancer (MCF7) cells with IC_{50} values of 56.1 and 76.5 μM, respectively, HIV-1 reverse transcriptase inhibitory activity with an IC_{50} of 31.7 μM, and mitogen activity toward mouse splenocytes. Because lectins are non-immunglobulin proteins that can agglutinate cells through sugar-specific binding sites, their mechanism of anticancer and antimicrobial action differs from that of other mushroom components, reviewed by Hassan et al.[94]

Kim et al.[8] also evaluated hot water (HWE), microwaved 50% ethanol (MWE), acidic (ACE), and alkaline (AKE) extracts of the fruit body (sporocarp) of *H. erinaceus* for their ability to induce apoptosis (programmed cell death) in U937 human monocytic leukemia cells. Cell culture, cell viability, cytotoxicity, flow cytometry, chromosomal DNA integrity, mitochondrial membrane potential, expression of pro- and anti-apoptotic proteins, and activation and inhibition of caspase assays were carried out to help define the mechanism of observed apoptosis. The aqueous and aqueous/ethanol extracts were active in all assays, whereas the acidic and alkaline extracts

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with similar proximate compositions were both inactive. The results of the bioassays with the active extracts are consistent with an apoptosis mechanism governing suppression of the cell proliferation pathway that involves activation of mitochondria-mediated caspase-3 and caspase-9 but not caspase-8. Proximate analysis of the freeze-dried mushroom powder showed that it contains high amounts of proteins, carbohydrates, and minerals. The described observations indicate that aqueous and aqueous/ethanol extracts of *H. erinaceus* that induced NO production in macrophage cells strongly suppressed the proliferation of U937 leukemia but not normal cells, by induction of apoptosis, suggesting that *H. erinaceus* may have therapeutic potential against human leukemia.

The novel HEG-5 polysaccharide-protein isolated from fermented *H. erinaceus* mycelia inhibited human gastric SGC-7901 cancer cells by promoting apoptosis and cell cycle arrest at the S-phase.[32] RT-PCR and Western blot analysis showed that HEG-5 decreased the expressions of Bcl2, PI3K, and AKT1 proteins and increased the expressions of caspase-3, caspase-8, p53, CDK4, Bax, and Bad, suggesting that caspase- and p53-dependent mitochondrial-mediated signaling pathways participate in the molecular events of apoptosis. The results imply that the novel polysaccharide might be a candidate for treating gastric cancer. The use of HEG-5 against human gastric cancer is expected to be successful in light of the fact that *H. erinaceus* mushrooms are reported to prevent gastric ulcers, chronic gastritis, and other digestive tract-related diseases.[95,96]

Chen et al.[97] isolated polysaccharides from eight mushroom species by water extraction and alcohol precipitation. The *H. erinaceus* polysaccharide exhibited the highest antioxidant capacity and strongly inhibited MCF-7 breast cancer cells.

Inhibition of Tumor Growth. Kim et al.[7] investigated antitumor effects of the four extracts of freeze-dried *H. erinaceus* in Balb/c mice intracutaneously transplanted on their backs with CT-26 colon cancer cells; HWE and MWE, with a higher content of β-glucans than ACE and MKE and determined using an assay kit, were active in all bioassays. GC-MS analyses showed the presence of 40, 27, 16, and 13 compounds, respectively, in the four extracts. Daily intraperitoneal (ip) injections of HWE and MWE for 2 weeks significantly reduced tumor weights by 38 and 41%. Tumor regressions were associated with changes in the following cancer biomarkers as compared to phosphate buffer (PBS)-treated control mice: 2.7- and 2.4-fold increases in cytolytic activity of splenic natural killer (NK) cells; restored nitric oxide production and phagocytosis (engulfing and destruction of cancer cells by phagocyte cells) in peritoneal macrophages to 95−98% of normal levels; 2-fold increase in released pro-inflammatory cytokines tumor necrosis factor-α, interleukin-1β, and interleukin-6 from macrophages; and 56 and 60% reductions in the number of blood vessels inside the tumor. The pro-angiogenic factors vascular endothelial growth factor (VEGF), cyclooxygenase 2 (COX-2), and 5-lipoxygenase (5-LOX) were also significantly reduced in mRNA and protein expression by tumor genes. Enzyme-linked immunosorbent assay of tumor cells confirmed reduced expression of COX-2 and 5-LOX (32 and 31%). Reduced COX-2 and 5-LOX expression down-regulated VEGF expression, resulting in the inhibition of neo-angiogenesis inside the tumors. It seems that induction of NK activity, activation of macrophages, and inhibition of angiogenesis all contribute to the mechanism of reducing tumor size.

Another study by Kim et al. investigated the antimetastatic activity of four *H. erinaceus* mushroom extracts using CT-26 murine colon carcinoma cells as an indicator of inhibition of cell migration to the lung.[6] The HWE and MWE extracts strongly elicited cancer cell death through apoptosis and inhibited metastasis of cancer cells to the lungs by 66 and 69%, respectively. The same extracts reduced the expression of matrix metalloproteinases MMP-2 and MMP-9 in cells and their activities in culture media. Urokinase-type plasminogen activator (u-PA), another extracellular matrix (ECM)-degrading proteinase, also showed decreased protein expression. In CT-26 cells, HWE and MWE down-regulated extracellular signal-regulated kinase (ERK), c-Jun N-terminal kinase (JNK), and p38 mitogen-activated protein kinase (MAPK) phosphorylations. The reduced phosphorylations seem to cause reduction of activity of the MMPs, thereby blocking migration and invasion of cells. Dietary administration of HWE and MWE reduced the formation of tumor nodules in the lung by about 50 and 55%, respectively, and prevented increases in lung weight caused by cancer cell metastasis.

It seems that HWE and MWE behaved as antimetastatic agents, targeting their upstream signaling molecules for mediating the expression of the ECM-degrading proteinases. Acidic and alkaline extracts were not bioactive. This study elucidated the mechanism of the observed inhibition of metastasis of cancer cells from one organ to another in terms of biomarkers known to be associated with prevention of metastasis, suggesting that inhibition of metastasis seems to be related to the composition of *H. erinaceus*.

Daily oral administration of three amounts (50, 100, and 200 mg/kg/day for 10 days) of *H. erinaceus* polysaccharides to 50 mice with S180 solid tumors reduced the increase in tumor weight from 100% for the control group to 68.5, 32.1, and 43.2% in treated groups, respectively, as compared to 18.5% for treatment with the cancer drug cyclophosphamide.[98] The reduced tumor weight gains were accompanied by increased serum levels of antigen-specific-γ (IFN-γ) and interleukin-2 (IL-2), suggesting that the antitumor effect might be associated with immunostimulation. A related study on the antitumor and immunoenhancing effects of polysaccharides is mentioned under Stimulation of the Immune System.

The cited studies suggest that *H. erinaceus* polysaccharides merit further study to determine if their anticarcinogenic activities in mice also occur in humans.

An interesting finding was that the replacement of aspartic acid/glutamic acid with asparagine/glutamine residues in ribonuclease from *H. erinaceus* resulted in a modified enzyme that inhibited the proliferation of a human leukemia cell line.[99] The authors suggest that the modified ribonuclease, which was expressed by recombinant methods in *E. coli* bacteria, might be of general importance for the production of anticancer biologics that are based on ribonuclease activity.

NEUROPROTECTION AGAINST NEURODEGENERATION

Cell and Mice Studies. A review by Wong et al. on the treatment of peripheral nerve injury by *H. erinaceus* concludes that the neuroprotective effects of *H. erinaceus* and other mushrooms in rodents need to be confirmed in human studies.[100] Here, we will be briefly examine selected studies on the neuroprotective effects of *H. erinaceus* in cells and animals and their potential impact on cognitive abilities in humans, which suggest that such mushrooms may help improve brain and nerve diseases in humans.[20,101,102]

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A systematic screening of 19 commercial *Hericium* strains revealed that the administration of fruiting bodies of *H. erinaceus* could attenuate (protect against) Aβ25-35-triggered damage in rat pheochromocytoma (PC12) cells, which are widely used to study neuritogenesis in vitro.[11] The observed effects were an increase in cell viability and a decrease in the release of lactate dehydrogenase; the described method could therefore be used to screen for the neuroprotective effects of mushrooms.

Nerve growth factor (NGF) is involved maintaining and organizing the function of cholinergic neurons of the basal forebrain complex (BFC) of the central nervous system, the neurons of which degenerate during the progression of Alzheimer's disease (dementia). Kawagishi et al. discovered and named the bioactive diterpenoids erinacine A, B, and C from *H. erinaceus* mycelia that stimulated NGF synthesis.[103] Phan et al.[104] cultured *H. erinaceus* mycelia and screened pure compounds isolated from the basidiocarps for neurite outgrowth activity using PC12 cells.

The use of HPLC, LC-MS, and ^{1}NMR facilitated the elucidation of the structures of six compounds from the basidiocarps: hericenones B−E, erinacerin A, and isohericerin. Hericenone E stimulated NGF secretion to a level that was 2 times higher than that of the positive control, by potentiating outgrowth in the PC12 cells, possibly mediated by ERK1/2 (extracellular signal-regulating kinases) and P12Akt (protein kinase) signaling pathways to induce neuronal differentiation and neuritogenesis.

A related study by Mori et al. examined the effect of four edible mushroom fruiting bodies including *H. erinaceus* on NGF gene expression in 1321N1 human astrocytoma cells (astrocytes).[105] The results show that *H. erinaceus* fruiting bodies, but not the isolated hericenones C, D, and E, promoted NGF mRNA expression in a concentration-dependent manner and enhanced the secretion of the NGF protein from the cells. Additional studies showed that the extract also enhanced the neurite outgrowth of PC12 cells in vitro and increased NGF mRNA expression in the hippocampus, but not in the cortex, of mice fed a 5% dry *H. erinaceus* powder for 7 days and that the stimulation of NGF production occurs via activation of the JNK pathway.

Another study by Mori et al. examined the effect of oral administration of *H. erinaceus* powder added to the mouse diet over a 23 day experimental period on the amyloid β(25−35) peptide-induced learning and memory deficits (cognitive dysfunctions) in mice.[106] Evaluation of the learning function of the mice by behavioral pharmacological methods (Y-maze and novel-object recognition tests) indicates that the mushroom powder improves the cognitive deficits induced by the amyloid peptide, suggesting that the powder might be a promising treatment for cognitive dysfunction. Additional HPLC studies showed that the powder contained the following three hericenones: C, 4.12 mg/g; D, 0.21 mg/g; and E, 0.49 mg/g.

The cited cell and mice feeding studies imply that because NGF levels decrease in the basal forebrain of Alzheimer's patients and in the frontal cortices of people with senile plaques, the results of the cell and mice feeding studies showing increased secretion of NGF indicated that hericenone E and the *H. erinaceus* powder merit further studies for their potential to prevent progress of dementia and senile plaques in humans.

Antifatigue Effects. A study by Li et al. examined the antifatigue activity of a polysaccharide extracted from *H. erinaceus* in a mouse model using the following criteria.[107] A control group and three test groups of mice were treated with the polysaccharide (50, 100, and 200 mg/kg for 28 days). The results of a force swimming test, a measure of fatigue, decreased blood lactic acid (BLA), serum urea nitrogen (SUN), and malondialdehyde (MDA) content, and increased tissue glycogen content and antioxidant enzyme activity suggest that an *H. erinaceus* component induced significant antifatigue activity and might, therefore, be of value in the field of sport nutrition.

Acceleration of Wound Healing in Rats. A Malaysian study found that the topical application of an *H. erinaceus* aqueous extract (0.2 mL of 20, 30, and 40 mg/mL) accelerated the rate of wound healing in rats.[108] The healed wounds contain fewer macrophages and more collagen, suggesting that the healing effect might be associated with the up-regulation of collagen formation and an increase in the tensile strength of the wounds. Because the livers and kidneys of the rats were not affected by the treatment, hematology and clinical biochemistry were within the range of control animals tested, and the highest dose of the extract (5 g/kg body weight) did not cause any toxicity, it seems that the mushroom extract is relatively nontoxic. The authors suggest the need to identify the bioactive component in the extract that has possible immunostimulatory potential.

Inhibition of Cell Senescence. Noh et al. evaluated the inhibitory effects of six compounds isolated from *H. erinaceus* on adriamycin-induced cellular senescence in human dermal fibroblasts (HDFs) and human umbilical vein endothelial cells (HUVECs).[109] One of the compounds—ergosterol peroxide—reduced senescence-associated β-galactosidase (SA-β-gal) activity that is increased in HUVECs treated with adriamycin, suggesting that one or more of the pure compounds might have the potential to treat/prevent age-related diseases in humans.

HUMAN STUDIES

Beneficial Effects on Depression and Anxiety. Nagano et al. investigated the effect of *H. erinaceus* on depression, menopause, and sleep quality using epidemiological-based questionaires.[110] In a randomized, double-blind, placebo-control trial conducted over 4 weeks, 30 females aged 41.3 ± 5.6 years were randomly assigned to either *H. erinaceus* or placebo groups and fed *H. erinaceus* cookies containing 0.5 g of the powdered fruiting body 4 times a day for 4 weeks or placebo cookies for the same regimen. The mean standard scores at the end of the test were lower for the *H. erinaceus* than for the placebo group, suggesting that orally consumed *H. erinaceus* has the potential to reduce depression and anxiety in women but this is by a different, and as yet unknown, mechanism from the NGF-enhancing effect by the mushroom extracts and hericenones mentioned above.

Improvement in Cognitive Impairment. Mori et al. examined the efficacy of oral administration of 250 mg tablets containing *H. erinaceus* mushrooms (prepared by air-drying harvested fruiting bodies at 60 °C overnight and powdering) to patients diagnosed with mild cognitive impairments.[111] A double-blind, parallel-group, placebo-controlled trial was used on two 15-person groups of 50−80-year-old Japanese men and women for 4 weeks. At weeks 8, 12, and 16 of the trial, the mushroom group significantly increased scores on the cognitive function scale compared with the placebo group. Following treatment, a comparison of 21 medical laboratory tests of the two groups showed no significant differences, suggesting that

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the mushroom tablets were safe and that their consumption improved mild cognitive impairment without serious adverse effects.

■ ANTIOXIDANT AND ANTIDIABETIC PROPERTIES

The antioxidant activity of *H. erinaceus* has been associated with the proposed treatment of several conditions. For example, the reduction of alloxan-induced diabetic neuropathic pain in rats following oral administration of an ethanol *H. erinaceus* extract was associated with a significant increase in the total in vivo antioxidant status (TAOS) in *Hericium*-treated groups, suggesting that the mushrooms might relieve diabetic neuropathic pain in humans.[112] The results of a related study on the antidiabetic and in vitro antioxidant activities of eight medicinal mushroom species seem to support the apparent relationship between antioxidant and antidiabetic effects.[113]

Indeed, it has also been shown that the ip administration of an aqueous extract of *H. erinaceus* (100 and 200 mg/kg body weight) to diabetic rats for 28 days resulted in a significant decrease in the serum glucose level and a significant increase in the insulin level and attenuated serum lipid profiles (disorders) as compared to control rats.[114] These findings were accompanied by increased activities in the antioxidative enzymes catalase (CAT), superoxide dismutase (SOD), and glutathione peroxidase (GSH-Px) and increased GSH (glutathione) and reduced malondialdehyde (MDA) levels in the liver, suggesting that the mechanism of the health-promoting effects seems to be the result of inhibition of ROS (oxidative stress parameters) known to damage the liver function. The specific compounds in the extract that may be responsible for the described effects are not known.

■ BENEFICIAL EFFECTS ON ISCHEMIA

Because ischemia (deficient supply of blood to organs) followed by reperfusion is a frequent clinical problem in cardiovascular surgery, Han et al. evaluated the effect of the preadministration of an *H. erinaceus* polysaccharide to mice by gavage at a dose of 300 mg/kg for 15 days on induced ischemia.[115] The results show that the treated mice had a significant decline in levels of blood urea nitrogen and serum creatinine and an increase in creatinine clearance levels compared to a renal ischemia−reperfusion control group. Additional studies with associated biomarkers (glutathione, malondialdehyde, antioxidative enzymes) showed that the polysaccharide pretreatment significantly decreased the lipid peroxidation level and increased antioxidant enzyme activities compared to the renal ischemia reperfusion group, suggesting that the polysaccharide prevented oxidative injury in the kidneys of the experimental animals and that it might help prevent acute kidney failure in humans.

■ ANTIOXIDATIVE AND HEPATOPROTECTIVE EFFECTS

As mentioned briefly above, *H. erinaceus* has been reported to have hepatoprotective effects, and these can be linked to its antioxidative activity. For example, a study by Zhang et al. of antioxidant and hepatoprotective effects of endo(intracellular)-polysaccharides from three fractions of *H. erinaceus* grown on the soybean processed product tofu whey found that the polysaccharide isolated from one of the fractions had strong in vitro antioxidant activity and protected mice against liver damage (assessed by observed changes in liver enzyme levels) induced by carbon tetrachloride (300 mg/kg body weight).[116] The beneficial effect on the liver is presumably the result of the potent antioxidant capacity of the polysaccharide, as determined by several different methods. The polysaccharides from the other two fractions were less effective. The most active polysaccharide could be used as a dietary supplement to prevent hepatic disease.

■ ANTIOXIDATIVE, ANTIHYPOLIPIDEMIC, AND ANTIHYPERTENSIVE EFFECTS

Feeding rats an exobiopolymer (molecular mass ∼ 40 kDa; protein and sugar contents of 8.8 and 91.2%, respectively) isolated from a submerged mycelia culture of *H. erinaceus* at a dose of 200 mg/kg body weight significantly reduced in a concentration-depended manner plasma total cholesterol (32.9%), bad low-density lipoprotein (LDL) cholesterol (45.4%), triglyceride level (34.3%), and the atherogenic index (58.7%) and increased good plasma high-density lipoprotein (HDL) cholesterol level (31.1%) as compared to the control diet.[117] The authors suggest the need to define the mechanism of the striking hypolipidemic effect.

Because oxidation of LDL might be a key factor in the pathogenesis of atherosclerosis, Rahman et al. investigated the in vitro inhibitory effect of *H. erinaceus* fractions extracted with different solvents on LDL oxidation and the activity of the cholesterol biosynthetic enzyme 3-hydroxy-3-methylglutaryl coenzyme A (HMG Co-A) reductase.[118] Of the solvent extractions, the hexane fraction showed the highest inhibition of oxidation and strongly inhibited the activity of the enzyme by up to 59.9% at 10 mg/mL. GC-MS analysis revealed that this fraction contained ergosterol and linoleic (9,12-octadecadienoic) and octadecanoic acids, suggesting that these components and possibly also a polysaccharide (not evaluated) might be responsible for the observed antioxidative properties of the mushroom extract and that the hexane extract might help to prevent oxidative stress-induced pathogenesis of atherosclerosis.

Because platelet aggregation in blood vessels causes vascular disease such as arterial thrombosis (formation of a blood clot) that may be accompanied by the formation of atherosclerotic lesions associated with myocardial infraction and stroke, Mori et al. evaluated the compound hericenone B isolated from *H. erinaceus* on collagen-induced platelet activity in rabbit and human platelets.[119] The results show that hericenone B exhibited strong antiplatelet aggregation activity by a mechanism that blocks collagen-induced platelet aggregation through the inhibition of arachidonic acid release in integrin $\alpha 2/\beta 1$ signaling, suggesting that it has potential to be used in antiplatelet therapy to prevent cardiovascular disease and stroke.

An in vitro study was conducted by Abdullah et al. on the inhibition of lipid peroxidation of buffered egg yolk and angiotensin converting enzyme inhibitory effects, an indicator of in vivo hypertensive activity, by water extracts of 10 culinary−medicinal mushrooms, including *H. erinaceus*. They found that *Ganoderma lucidum* mushrooms, with the highest phenolic content (63.5 GAE/g extract), exhibited the highest angiotensin converting enzyme inhibitory activity (IC_{50} = 0.05 mg/mL),[120] whereas the corresponding values for *H. erinaceus* were much lower: 10.2 GAE/g and IC_{50} = 0.58 mg/mL. There seems to be a good correlation between total phenolic content of the 10 mushroom varieties and their antioxidant capacities as determined by different methods. To determine whether

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mushroom extracts can serve as protective agents against hypertension, however, the in vitro results need to be reproduced in vivo, in particular in humans, as is the case for many of these studies.

STIMULATION OF THE IMMUNE SYSTEM

Here, we will briefly describe reported studies on the stimulation of the immunes system in various animals and also the effect on dendritic cells by *H. erinaceus* mushrooms and associated polysaccharides.

Wang et al. compared the antitumor and immunoenhancing effects of polysaccharides with molecular weight > 1×10^5 kDa isolated from *H. erinaceus* and *H. laciniatum* mushrooms cultivated in a shaker at 25 °C for 25 days.[121] The results showed that both polysaccharides had significant antiartificial pulmonary metastatic tumor effects in mice, with the polysaccharide from *H. erinaceus* being more effective than that from *H. laciniatum* and that both enhanced the increase of T cells in the mouse model at the same rate, indicated by higher numbers of $CD4^+$ cells and macrophages in the test than in the control group. The results suggest that the antitumor effects might be associated with the immunoenhancing activities of the polysaccharides.

Supplementation of the diet of the olive flounder fish (*Paralichthys olivaceus*) with 0.1 and 1.0% *H. erinaceus* mushroom protected the fish that were infected with the parasite *Philasterides dicentrarchi* against mortality via enhancement of the innate immune response, as indicated by cumulative mortalities during the 4 weeks of 45 and 30%, respectively, compared to 90% in the control group, suggesting that the mushrooms modulated the nonspecific defense mechanisms against the pathogen.[122] The protection was associated with an increase in scuticocidal activity, enhanced respiratory burst activity of phagocyte cells, phagocytosis, and lysozyme activities in plasma. The authors suggest that, because of their biodegradability, mushroom diets merit further study for their beneficial effects in aquaculture.

A related study evaluated the effect of particle size (100−150, 74−100, and <74 μm) of dried *H. erinaceus* powder on the immune response and disease resistance against the pathogen *Vibrio alginolyticus* in white shrimp (*Litopenaeus vannamei*).[123] The results show that shrimp mortality was particle size dependent, with particles of <74 μm inducing a significantly greater effect on the disease resistance than those >74 μm, as indicated by a significantly lower mortality of the shrimp during the 28 day test period with a diet containing the mushroom powder at a concentration of 0.2 μg/g shrimp. The protective effect was accompanied by an increase in phenol oxidase activity, respiratory bursts, superoxide dismutase activity, and glutathione peroxidase activity, parameters usually associated with destruction of ROS. It seems that the smaller particle size improves the bioavailability of the dietary mushroom powder.

Dendritic cells are antigen-presenting cells that act as mediators of peripheral immune tolerance and help maintain immune homeostasis.[124,125] An evaluation of several biomarkers associated with the immunostimulatory effects of polysaccharides isolated from *H. erinaceus* mushrooms on murine dendritic cells showed that treatment with 50 μg/L of a 20% β-glucan-containing polysaccharide extract induced activation and promoted morphological changes, reduced endocytosis (incorporation of the polysaccharide into the cells by phagocytosis), increased the expression of surface molecules that are important for antigen presentation, and stimulated the secretion of cytokines that promote immune responses,[126] suggesting the potential value of the polysaccharides as an immunostimulatory food additive and adjuvant to enhance the immunogenicity of vaccines.

OUTLOOK

This overview covers an area of food medical science of current importance to mushroom growers, consumers, and food and biomedical scientists. Although the described results of the extensive, worldwide studies on multifunctional health properties of *H. erinaceus* in cells and in rodents, and to some extent in humans, need to be further confirmed in human nutritional, clinical, and epidemiological studies, a number of suggested research needs on the chemistry and numerous health-promoting aspects have been mentioned in the text. Because *H. erinaceus* species seems to lack toxicity in rodents and humans, nutritionists, pharmacists, and physicians should consider recommending these mushrooms to their clients and patients. The apparent exceptional beneficial properties will, hopefully, induce mushroom growers in the United States and elsewhere to produce *H. erinaceus* mushrooms for sale in stores. Because a large number of other *Hericium* species exist in nature, it would also be of interest to compare the described results on the composition, nutrition, and health benefits of *H. erinaceus* to phylogenetically related species. The suggested studies might lead to the discovery of new, highly bioactive *Hericium* species with high health-promoting potential. We anticipate that these efforts will contribute to the improvement of human nutrition and health.

It is also relevant to note that a meal consisting of *H. erinaceus* and other foods such as tomatoes that contain bioactive lycopene and tomatine[90] or pigmented rice that contains bioactive γ-oryzanol[127] will contribute multiple types of health-promoting properties to the diet that might induce additive or synergistic beneficial effects after consumption. This aspect also merits study. Finally, we expect readers to also benefit from our earlier related reviews on the health benefits of other food categories and their bioactive compounds that complement the present review.[77,79,87,91,127−131]

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Notes

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ABBREVIATIONS USED

ACE, AKE, HWE, MWE: acid, alkaline, hot water, and microwave *H. erinaceus* extracts
CD-4: helper T cells that promote adhesion to antigen-presenting cells in response to infection
Bad, Bax: pro-apoptotic proteins
Bcl2,: apoptotic protein
BFC: basal forebrain complex
BUN: blood urea nitrogen
COX-2: cyclooxygenase-2
BLA: blood lactic acid

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ECM: extracellular matrix
DC: dendritic cells
EPS: exopolysaccharides
ERK: extracellular signal-regulating kinase
FB: fruit body
FT-IR: Fourier transform infrared spectroscopy
GABA: γ-aminobutyric acid
GAE: gallic acid equivalents
HDFs: human dermal fibroblasts
HEG-5: *H erinaceus* glycoprotein
HUVECs: human umbilical vein endothelial cells
HPSEC-MALLS-RID: high-performance size exclusion chromatography with multiangle laser light scattering and refractive index detection
IC_{50}: concentration of test compound that inhibits 50% of cancer cells
IFN-γ: antigen-specific-γ
IL: interleukin-2
IPS: intracellular polysaccharides
ITS: rDNA internal transcribed tracer
JNK: c-Jun N-terminal kinase
5-LOX: 5-lipoxygenase
MAPK: mitogen-activated protein kinase
MIC: minimum inhibitory concentration
MDA: malondialdehyde
MMP: metalloproteinase
MRSA: methicillin-resistant *Staphylococcus aureus*
MTT: methyl tetrazolium cell viability assay
NF-κB: nuclear factor kappa-light-chain-enhancer of activated B cells
NK: killer cells
NGF: nerve growth factor
P12Aκt: protein kinase
RT-PCR: reverse transcription polymerase chain reaction
ROS: reactive oxygen species
TAOS: total in vivo antioxidant status
u-PA: urokinase-type plasminogen activator
VGEF: endothelial growth factor

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